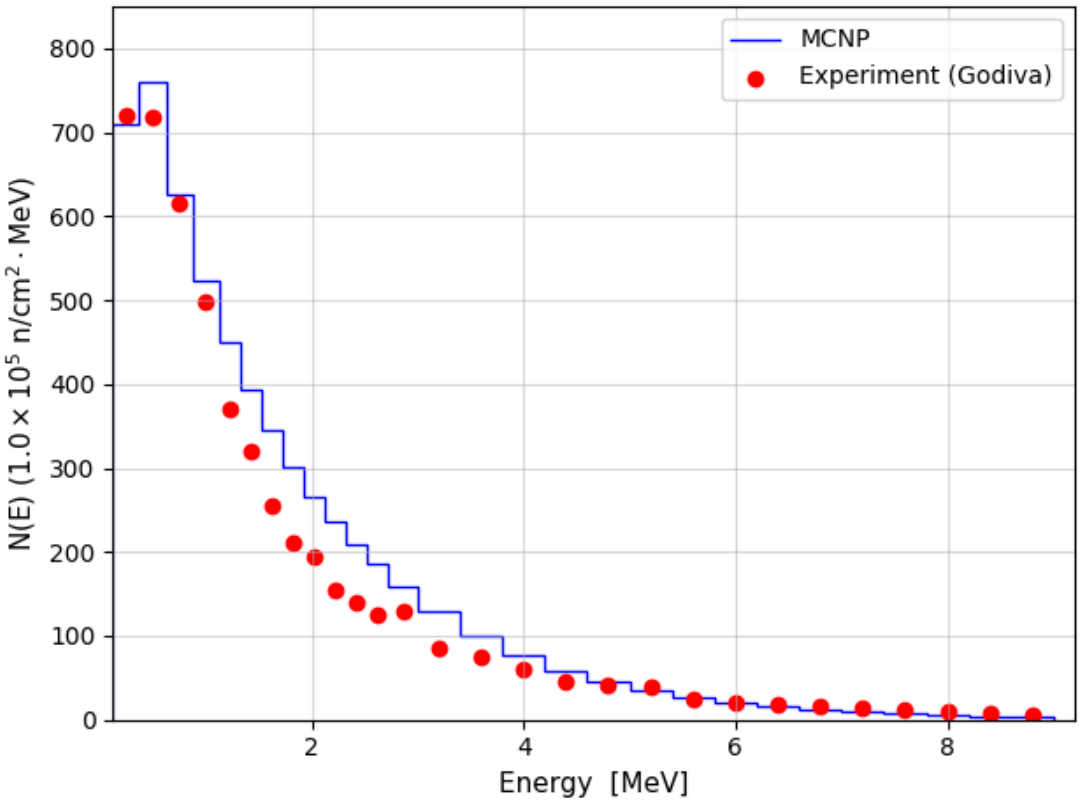

Figure 1: Comparison of Leakage Current for Experimental vs MCNP Simulation of the Godiva Assembly

From Fig. 1, we see good agreement between the experimental and MCNP results. MCNP generally over-predicted the flux until roughly 5 MeV where the MCNP and experimental results agreed much more closely.

At the 2000th batch of the simulation, I had a keff of 1.00440. This is very close to the desired critical value of 1.00000. There are myriad potential reasons for the discrepancy, for example: poor cross-section data and these results were measured on very old equipment. However, it is important to see, even in such a small and simple model, there will always be a discrepency between the simulation and the real world.